Solution Requirements

Date	16 October 2026
Team ID	
Project Name	Personalized Travel Itinerary Planner
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	*Secure user login and signup via Supabase Auth supporting Email/Password and Google OAuth with encrypted JWT session tokens.*	High
2	Authorization levels	*Role-Based Access Control (RBAC): Traveler (create, edit, and export personal trips) and Admin (manage destination datasets and vibe rules), enforced via Supabase Row-Level Security (RLS).*	High
3	External interfaces	*Integrates with Google Gemini 2.5 Flash API (AI prompt synthesis), Supabase PostgreSQL*	High

		(cloud storage), and external calendar apps (Google/Apple Calendar).	
4	Transactions processing	*Atomic CRUD operations for trip creation, saving itinerary days/activities, and updating reordered route sequences with local offline caching.*	High
5	Reporting	*Generates interactive day-by-day itinerary timelines, iCalendar (.ics) event files, downloadable luxury PDF travel dossiers, and raw JSON export.*	Medium
6	Business rules, etc.	*Enforces travel constraints: activity counts strictly based on pace (Relaxed: 2/day, Moderate: 3-4/day, Fast: 5+/day); strict daily budget capping; and 2-opt TSP route sorting to eliminate backtracking.*	High
7	Compliance to laws or regulations	*Complies with GDPR and privacy standards: user consent for preference storage, complete data deletion on request, and HTTPS/TLS encryption.*	Medium
8	*Other*	*Real-time dynamic vibe tuning sliders (+40% Adventure, Sunset & Chill Mode) allowing instant activity recalculation and schedule swapping.*	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria

1	Performance & Speed	*Rapid UI page rendering, fast AI schedule generation, and low-latency route sorting algorithms.*	*• Page load time < 1.5 seconds* *• AI itinerary generation < 3.0 seconds* *• Route TSP calculation < 250 ms*
2	Scalability	*Stateless FastAPI REST backend deployable on Render with database connection pooling to Supabase.*	*• Supports up to 5,000 concurrent active users without degradation.*
3	Security & Data Privacy	*End-to-end data encryption in transit and database isolation ensuring users only access their own trips.*	*• HTTPS/TLS 1.3 encryption* *• 100% Supabase Row-Level Security (RLS) policy coverage.*
4	Reliability & Availability	*Continuous system uptime with graceful offline fallback if the Gemini AI API experiences downtime.*	*• 99.9% uptime per month* *• Fallback cached recommendations load in < 500 ms on API timeout.*
5	Usability & Accessibility	*Responsive glassmorphism interface functional across mobile, tablet, and desktop screens.*	*• Complies with WCAG 2.1 AA standards* *• Mobile responsiveness score > 90/100 on Google Lighthouse.*
6	*Other*	*Modular, decoupled architecture separating Vite React frontend and FastAPI backend for zero-friction containerized deployment.*	*• Full local setup via npm and uvicorn in < 5 minutes.*